P02

Topological Foundations of Kirchhoff Current and Voltage Laws

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Domain	DOM_PHYS

Abstract

We formulate the conservation of current at vertices (cutset law) and potential continuity around cycles (cycle law) using incidence matrices of directed graphs, laying the groundwork for network graph theory.

Abstract quoted from G. Kirchhoff et al., Annalen der Physik (1847).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?